Homology and convergence in β-galactosidases.

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Abstract

The structure of orthologous group genes are presented as a proposal that can be used to compare and recognize particular conserved and unique domains, to define by comparison a convergent and/or conserved homologous features. Their constitution can be based on comparison, functional analysis or a biochemical character as a shared reaction.

The β-galactosidase reaction involved in at least four know EC classification is distributed among eight orthologous groups in KEGG. These orthologous are integrated in this comparative study, where by phylogeny and comparative structural homology allow to establish shared and differential elements of different KO groups. The methodology described in this paper allow to establish at least three different paths that bring a complete diversity and distribution among the prokaryotic domains.

Keywords— homology, structural comparative, beta-galactosidase, orthologous groups

Introduction

The study and proposal of orthologous groups have been one of the most frequent practices by molecular biologists and even previous to the development of comparative genomics and gene family analysis and studies. According with Caporale [1], is possible to propose a group of genes to study and recognize shared and unique features based in particular and differential arguments.

Actually, in recent studies is more frequent to define a group of orthologous genes applying comparative genomics into a group of organisms[2]. Defining a group of genome sequences

to compare and establishing a value of similarity among all the analyzed sequences, enable to define a group of shared genes [3]. These genes, according with the grouping proposal and previous reference might be present into the genomic structure of a hypothetical common ancestor from all the studied sample [4].

There is an alternate way to establish orthologous groups: comparing metabolic function and participation of certain gene products in the organism' pathways and infer a functional homology in two or more species. The case of the KO groups [5] is defined in similar way. The KO groups are defined by the comparison and homology finding into its biochemical information. This data is combined with the participation of the same gene products into molecular metabolic networks who define hierarchies and modules in their function and representation [6].

In most of these studies the prevalent hypothesis is to define that a particular element of metabolism is conserved in all representatives who have the sequence and if this catalytic reaction has a particular shared evolutionary history in all analyzed species. One example that actually is being studied is all the *lacZ* β-galatosidases [7]. In other analysis is possible to define if the structural comparison allow to define how is the relationship among the different subgroups presented inside our analysis and finally compare into a wide open system if is possible to recognize since particular or partial bias and shared properties among one or different groups.

The β-galatosidase enzymes are well known metabolic persistent elements involved in the metabolism of galactose and lactose. β-galatosidase enzyme were primarily described as a product of the *lacZ* gene [8]. This enzyme is an hydrolase and can recognize and degrade reduced ends of beta-galactosides. Their functional structure can be recognized as a homotetramer whose primary chain are form by five structural regions and is conserved into most of representative bacteria involved into capabilities enable to degrade even synthetic compounds only with a shared formulation or particular galactosides [8]. However the prevalence of this reaction is not unique. At least in three genes: (*ebgA*, *bgaB*, and *lacA*) are reported with the same EC

number (consulted in the same KEGG database) [5]. The early comparison and recognition of β-galatosidases enable to consider that this might be the result of early diversification of these genes that can be the result in cellular context in Eukarya or an specific duplication and divergence only into certain phylogenetic groups [9, 10]. However, this possible condition only brings more questions about this topic. Are all these three genes with a similar reaction could be related? Do they have a shared similar tertiary structure and that is sufficient to vinculate them into a shared pattern?, and does is possible to recognize whose are the early model of this reaction.

The knowledge of tertiary structure proteins have been showed as a support by different methods to identify structural/functional characteristics of proteins and could be used for the detection of remote homologues [11]. The main factor that facilitate the use of tertiary structures is to make a practical comparison among different crystals, that enable to avoid several silent changes present in primary structures, that can be the product of particular phylogenetic changes[12].

One choice to infer possible folding homologies in one or more cases is to use specific web server services based in two measures RMSD and TM-score.

When two structures are compared into a three-dimensional space only to recognize the possible equal positions, their equal difference between them can be expressed in terms of a root-mean-square deviation (RMSD). The RMSD measure is intuitive, and it fulfills the definition of a metric. Alternatively, interatomic distance matrices are measure by pairs and can be compared by yielding a difference measure in distance space. This measure are specialized and applied not only as pairwise comparison, but can be applied into a group of structures. A useful measure for the "spread" for a complete group comparison is to create an average RMSD of all pairs of structures. This method is based in the Kabsch's formula for calculating pairwise basic comparison. The resulting values can be expressed as a covariance matrix which is built by a isotropically distributed ensemble (IDE) [13]. The calculations around the IDE covariation matrix integrates a result that enables to bring addition in the comparison of multiple models

and recognnize ~~whch~~ structure is more close to the average value of all the compared group. RMSD is a metric that allow to propose an structural similarity, however this globally optimal transformation minimizes the distances between all superimposed atom pairs and sometimes can be dominated only by a small set of divergent atoms in loop regions. As we can consider, according with recent studies [14] Furthermore, RMSD is not only dependent on the overall goodness of fit but also dependent on the length of the proteins. Is in this case that we have decided an alternate measure to compare due to the β-galactosidase has an average short average amino acid sequence. The reference of one of the studies revisited to comprehend this comparison was the work of Cuneo et al. into the comparison of a thermophilic enzyme in tertiary structure against a mesophilic [15] in in this work where is possible to recognize a shared folding based of the docking and ~~en-palming~~ of almost the 85% of the alpha carbons of the backbone.

The TM-score by counterpart TM-score uses a variant of the Levitt–Gerstein (LG) metric (Gerstein and Levitt, 1998) that gbring a length - independent metrics and limits the impact of divergent pairs of compared atoms in superimposed structures. For the case of TM-score, there are no simple relationship between the covariance matrix and the optimal transformation that maximizes LG. In comparison the TM-score applies different subsets of atoms are superimposed using the Kabsch algorithm and the LG score measured over the entire protein. After sampling a large number of subsets, is possible to have an approximately optimal superposition. TM-score algorithm is much slower than the calculation of global RMSD because its impolication to calculate multiple tries and events of superposition [14].

The TM - score, as a counterpart measure, allow to compare the configuration and to evaluate additional branches in the configuration ion the enzyme structure. Xu and Zhang [16], proposers of this work, have discussed about this measure and developed a mathematical approach only to define that a TM-score greater than 0.5 is suitable to consider as the same fold into pairwise or multiple alignments for proteins .

To compare structural elements enable to infer conserved regions and properties and in

this particular case of the β-galatosidases groups let us try a comparison where is possible to compare short chain peptides and in later events structural conformation. In certain groups and gene families the β-galatosidases can be recognized as homotetramers but this model changes in several structural homologues [8]. Additional to this comparison, the use of previously reported motifs can be a further filter to recognize particular elements suitable to enrich the structural comparison.

The identification of homologous groups have an outstanding practical importance for the visualization and to recognize if their relation remain conserved after their proposal. The use of a search algorithm to compare with a well curated database is suitable to have additional information to infer, recognize and if is the case avoid an element for the grouping proposal [17]. One of the most important databases who has upgraded the analysis since alignment to structural comparison is the use of wellknown motifs. The main reference to the most complete database is Pfam.

Pfam is one of the most used family databases, built in many years of work by domain experts with extensive use of manual curation. This approach is generally very accurate, and practical because only working with primary structure sequence, it can recognize an alignment and a possible function of the sequence region [18, 19]. The use of this tool is the last element presented to improve since our comparison to the proposal of robust and bigger groups that look for establish groups of homologous elements who could share and infer conserved elements and functional properties in related-β-galatosidases. The objective of this study is recognize if is possible to fuse different β-galatosidases KO groups by combining conserved motif, matched tertiary structure and recognize possible conserved regions in the structure of proteins and if is possible beyond the EC number, to infer possible phylogenetic regions among the orthologous groups.

The enzymes related into β-galactosidases although diverse might be possible to link them into a common evolutionary path. The diversity shown even in particular phylogenetic bacterial groups are a good opportunity to recognize even a recent diversification or some elements who

show possible conserved elements and might let infer the *lacZ* diversification or the convergence and substitution in different phylogenetic representatives.

Methods

KO groups selection and Sequence database for each group

The KO orthologous groups proposed by this study were found using the "search keyword" section from the KEGG webpage [20]. The orthologous groups where selected with the keywords "beta-galactosidase" expression. The complete set of selected orthologous groups were consulted each one of them, capturing the properties describe in the KEGG general sheet from each link, defining their EC number, their sequence representation and if is possible to relate a particular gene with its group. All the data is presented in Table 1.

Additionally for each KO group, a set of sequences with at least one representative sequence per organism was collected manually. Because of the incidence of duplicated reports, it has taken the decision to only use one sequence per species in the later alignment and phylogenetic tree.

All the amino acid sequences were copied using the links in the KEGG webpage [20] to have the complete FASTA format with header referencing the KO group and the three letter code to identify the organism. These files containing all the sequences were aligned with MUSCLE [21]. This process of alignment enable to recognize which sequence is more conserved for all the KO and can be used for three purposes: *a)* the search of homologous structure corresponding to the KO group; *b)* the basal alignment and phylogenetic signal represented that was used in Figure 1; *c)* and the search in the Pfam database.

Search in PDB database

For each of the eight orthologous group an amino acid sequences were selected. The MUSCLE alignment enable to infer which are the most conserved sequence for each KO group [21]. This sequence were used to look for a particular homologous structure using the "Sequence Similarity Search" in the PDB webpage [22]. Using the conserved sequence for each KO group made possible to link each KO group with a particular PDB crystal model. The result of this search is present in Table 1.

Search in Pfam database - Combining motifs with structure

After find the PDB similar crystal corresponding to the KO group both, the sequence of the alignment and the primary structure for the PDB entry, were used to make a motif search in Pfam [19]. This method is used to corroborate that the Pfam motifs are identical and their crystal and in the KO query sequence. Only those motifs with values lower than e-10 are acceptable to work with, this is supported by previous report by Russo *et al.* [23] where is recommended to make multiple analyses of sequences or review the particular alignment. The data obtained is used into a figure where crystal and the KO are presented. Additionally a modification of the motif recognition is added, because the comparison of KO groups involving β-galatosidases might be a topic where the common ancestor of all those enzymes might be separated from a great evolutionary time and not a recent duplication and diversification event, because the differential structure and differences present in at least three great groups.

To evidence and integrate all Pfam data with the vinculated structures analyzed, it was used as a guide the PyMOL program [24] to select and follow the regions that are reported in Pfam and represented in color patterns by using since the selection to the alternate color patterns to differentiate these elements

First phylogenetic tree guide and distribution for each KO group

After recognize the most conserved sequence present for each KO group, each one of these representative sequences were copied and saved into a unique FASTA file that was aligned, with MUSCLE too [21], to use it as seed in the construction of a phylogenetic tree who resemble the probable relation among the KO orthologous groups. The tree was produced using the IQ-tree software [25]. This program was selected because is viable to realize a substitution model for the alignment and a bootstrap using samples with different sequence numbers and variable sequence length. This tree, presented in the Figure 1

Alternate method - Shoot.bio orthologous sequences

In order to have an additional vision that not only in KEGG sequences and the alignment proposal to obtain conserved sequences ~~is~~ found an alternate option that bring an automatic set of orthologous sequences based in Tree of life research project. The labor and background of SHOOT.bio is to recognize how is the the understanding and recognition of orthologs present now for all the completed genome coding projects [26]. Is in this effor~~d~~ where is possible to combine the Homology trees and clusters obtained with the gene actual divesity defined and structurized by elements whose are selected by the search HMMER algorithm and resulting in profiles and sets of probable homologous sequences [27]. The SHOOT.bio server webpage [28] enables to recognize suitable orthologous ser~~q~~uences whose are referenced in profile groups and homology groups with a diverse sample and representative value of homology.

The conserved and consensus sequence obtained from the MUSCLE alignments using the manually selected KO sequences over the Seaview software [29]. The obtained consensus and conserved sequences were used as a query for the SHOOT.bio system. These sequences were aligned in the same way, using the MUSCLE algorithm, and the result this analyzed was used by IQ-tree software, realizing since a bootstrap of 1000 repeats and a MODULETEST only to recognize and arrange into a more robust way all the sequences and probe that all these elements are informational [25]

The final consensus tree obtained from IQ-tree was read by Figtree, a software who can drive the image and configuration of the phylogenetic tree [30], cotract the overrepresented sequences and remark by different colors all those orthologous groups that share a taxonomy arrangement.

Results and Discussion

0.1 Crystal structure and its link with KO orthologous groups

The primary phylogenetic tree is used to have a first approach to compare and recognize probable KO groups closer or similar to compare and define the pairwise and multiple crystal comparisons.

For these cases the proposal obtained like a probable result are both K01190 whose sequences are directly related to the original *lacZ* gene and is compared with the K12111 orthologous group who are linked with all the *ebgA* genes. Here the main comparison resemble since the same Pfam motifs and seem to be the same fold, the shared signal of these group might be clearer in Figure 6 when almost all the sequences of both orthologous groups resemble the same phylogenetic grouping and signal

The second group to compare are the possibility to recognize that K12309 and K25543 that group all those sequences of the gene *GLB1* and *GLB1L* whose is a β-galactosidase enzyme with specific function in animals and specifically in leucocytes and fibroblast in human and mammal context. The remaining group suitable to compare, is the K12308 whose sequences are from prokaryotic origin (Bacteria and Archaea).

The third group resemble all the alternate branches that are not making a taxonomical clustering or its original value in the phylogenetic tree might not resemble into a clear homology and probable comparison. These groups are K01220, K18579 and the K01224 whose proposal of orthologous link for *lacG* gene, an uncertain coding sequence and *6GAL* gene respectively. Important is to know that beside the correct correspondence of previously reported genes the

reaction relate to the group K18579 with K01224 because in both cases these proteins can relate in their similar substrate: galactan and arabino-galactan galactosidase respectively. The remaining group K01220 is only related by an alternated substrate a phosphorylated version of galactoside. Is in this comparison that beside of the differential substrate these two comparison can not be subtracted as the configuration of an homologous sequence.

Linking consensus sequences with tertiary structure models

KEGG made possible to recognize at least nine different orthologous groups of $\beta-galactosidase$-related. One group were eliminated and used as an external group. The KO group K12112, whose EC is not characterized, their conserved sequences are only involving enzymes implied in the biofilm construction (reported in Uniprot) [31] and their differential length and Pfam motifs allow to avoid it as a functional beta-galactosidase, but still have a remnant signal who can be used as an external group, the Pfam motif reported as DUF386, that is a motif similar to those reported as Glyco-hydro in other orthologous groups. The actual result of the alignment in the different orthologous enable to infer a consensus sequence, to recognize element who are close to a shared signal for all the elements and use it as a wild card in the search of crystal structures and a preview phylogenetic relation among the KO orthologous groups showed in the Figure 1.

For the case to search a homolog tertiary structure model, although the search in structure and recognition might be only a search among tertiary structure models, a particular element is present in this search: the mmseq2 algorithm [32]. The usage of the mmseq2 element is the key to recognize into a search into massive databases and identifying homologous sequences and elements whose components are weighted into a structure that allow to link since crystal structures to functional motifs. The implementation of this search algorithm enable to find into a non-exhaustive search two elements that a are vital for our orthologous comparisons.

Pfam recognition and a perspective in homology

In this search, we have recognized that results with values greater that a 1e-15 might present a disperse proportion of identities of even similarities. Is vital to automatize in later studies only to corroborate that every KO group can also generate a cluster of homologous sequences who share the motif presence into an optimal value of conservation. The values for each KO orthology group is presented into Table 2, where is possible to corroborate the feature and character of each KO group and Pfam motif.

Primary result in the phylogenetic relation with orthologous groups

The first approach to relate a set of KO orthologous groups bring some elements to discuss. Although the proposal are sustained with an adequate bootstrap, to only establish a particular single sequence to represent might be an incomplete vision of the comparison. However this proposal might be coherent with the later structural comparisons in the three sets of sequences. If this approach is used only to recognize a first cluster to compare, that could be a decision well supported because the later comparison might complete about how the relation in all the KO groups and structural comparison can initiate. This kind of combined analysis bring us particular features that can be practical. Thank to *Figure 2* is possible to consider that the most diverse KO groups are K01190 and K12308, because the incidence presence of the genera analyzed and grouped in them, the number of sequences linked to each orthologous group but even more, their particular grouping in at least five and three different denomination of genes, this can be detected in Figure 2 but more in Table 1 and 2. Both groups involving in the same reaction and even referenced in previous literature as categorical β-galactosidases, however only one is related with lac operon while *bgaB* is reported independent from this structure is quite independent and different. Although is necessary to make further search of *bgaB* genes, one might consider a particular and unique structural β-galactosidase independent from a well

structured regulated gen. The *bgaB* have been reported in extremophiles and organisms of free-living lifestyle. The *bgaB* gene resemble as a gen with a particular phylogenetic signal group, that could explain their prevalence in Firmicutes [33].

The other case presented as a brother group one from the another is the K01190 group with the K12111. The resemblance of the lacZ with ebgA gene homology have been previously reported [34]. Their similitude and its conserved signal that goes upt to 50% of similarities could reference them as genes with the same origin and a common ancestor. However, this analysis needs further studies only to corroborate that the appearance of this genes might be involved in the shared signal evolution of lac operon. This couple of groups can be consider real homologous or convergent lineages into a single structure and function.

The second arrangements to compare is only references into eukaryal diversification groups K25543 an K12309 these two sequences enable to recognize the probable homology of genes GLB1 and GLB1L [35]. The role of these enzyme is completely defined, as some references establishes lysosomal biochemical tool involved into immune response and expression in animal leukocytes or fibroblasts. Their particular mutations and pathogenesis reported in medicine have allowed a complete study because not only in immunology is significative in their expression, additionally this enzymes are vital for the organization of pluricellularity because their absence or in cases of deletereous mutation can induce a phenotype of cancer [36]. This result brings a particular way to recognize that even in the history of animal evolution some β-galactosidases originated from the eukaryotic lineage remains as a particular path.

The remaining branches involved in β-galactosidase reaction K18579, K01224 and K01220 are not linked among them like the previous clusters of two or three groups, even the clustering value of bootstrap do not correlate them among these three remaining groups and are showed in Figure 3. The structure of all K01220 - *lacG* genes can remain as a particular fold that have particular shared properties that at least allow to recognize them as similar structures among them but only partially and evenmore, the properties linked in this analysis is affected by the alpha helices length and orientation.

The comparison according with the first phylogenetic approach

The structural comparisons showed in Figure 3, 4 and 5 seem to be different scales of comparison that can resemble the different scale of similitude where according with the results is plausible to infer a real homology.

The higher value of similarity proposed as an element that is supported even for the Pfam shared motifs are the comparison of K12111 (PDB:6S6Z) and K01190 (PDB:3J7H) the similarity that resemble those structures are completely suitable to define as two structures with the same fold corresponding with the argument of Xu and Zhang [16]. The differences of folding are partially visualized in the turns present in the *Glyco hydro 2 N* region that can be plausible in the recognition and promiscuity present for both functions. Another suitable explanation of the additional turns and those with an alternate orientation are the difference among the interaction of the complex that K01190 can resemble with other enzymes involved in the metabolism of lactose and galactose [37].

The second result that is detected is presented in Figure 4 who integrate the comparison of the K12309 with K25543 and in later stance K12308. Is in this analysis that even K12309 do not reach the homology fold that is not suitable to recognize and remain only as a close similarity a possible recognition of the functional Pfam motif might be responsible of this partial similarity. Using an alternate measure of similarity, we can sustain this proposal after analyze the similarity is higher in the active site. The remaining beta sheets domains are the differential elements because the volume and orientation are different among the two crystal models.

A way to recognize that not all the functional enzymes who looks similar are homologous are the later group of K12308 that is presented by the PDB:5E9A crystal structure. Is in this case from the absence to recognize Pfam motifs and the comparison of this structure with previous to groups where we can infer that although is a *bgaB* and *lacA* structure although the reaction is identical, the structure do not resemble similar element in any of the other groups. A possible reason to infer this difference can be since the enzyme phylogenetic origin as a prokaryotic enzyme that is compared directly with both eukaryal structures

The KO groups K12309 and K25543 are representing eukaryal genes important to the activity in multicellular interaction and immune reaction, even these ~~to~~ genes products and structures are completely separated although resemble a single similarity who are represented by the catalytic site and Pfam motif. That could only be explained as although their origin can be proposed as a duplicated gene in early history of eukaryotic evolution, this event seems to have a long number of changes and derivation of several regions such as the two beta sheet regions and even in the catalytic region a differential conformation of α.helices. It is in this comparison where is only viable to represent and recognize these three compared groups as three different path in the degradation of galactosides, with a simple recognition that need additional probes to infer all the changes that are present in K12309-3THC and K25543-4MAD structures is possible to propose a long distance ancestor in Eukarya evolutionary history. The remaining recognition of K12308-5E9A remains still as a crystal convergence in the prokaryotic context.

The only other homology that still is partial that is still in the verge of the homologous folding is the recognition of K01220-1PBG structure with K18579-6IUJ structure, the main reason is again the Pfam motif region recognized by the *Glyco hydro 30-2* motif that seems homologous in only a partial region involving *Glyco hydro 1* motif for K01220. This comparison enable to link a α-region where is plausible to recognize and link it with even the cofactor interaction. Although is not sufficient to only propose the homology for all the structure. Another element who make possible to avoid a clear homology are the beta sheet who is a region that extends from the central part and even brings additional feature in the polymerization properties and chain interactions. These groups are separated to even the representation of the homologous sequence. For this case K01220 that is the most recognized *lacZ* β-galactosidase remains as the most conserved group, but even in the last figure is only clear to represent it as a sole component with no additional KO group represented by homologous sequences. A different case is the alternate comparison of K01224-1R8L with K18579-6IUJ that resemble even a twilight result with only 0.43 of TM score, a differential value that is showed in the final phylogenetic tree.

About the composition and arrangement of orthologous group from Shoot.bio

The phylogenetic tree allow to recognize particular feature and a support of the comparison at structural level, presenting the same result of homologous K12111 and K01190, representing gene products presented mainly in the prokaryotic diversity. This result support the early recognition of both *ebgA* and *lacZ* genes

The recognition and comparison of structures representing and additional not probed comparison, the K01224 with the KO group, a comparison of *6GAL* gene with the proposed KO group K18579 not related into a particular gene permit to consider all those combination in the structural comparison, additionally the shared distribution of both groups resemble the possibility to support and compare in later studies only to recognize if the sequences remains as homologous. The differential distribution for these two groups might be possible to recognize if these elements present and homology and additionally a possible step in the horizontal enrichment from Bacteria to Eukarya, again this study is still present as a probable follow-up of the first approach relation among KO groups. This coherent result is even coherent with the first approach tree from figure 2 that resemble in different way how these elements would be near.

Finally what is remains in orthologous groups like K25543 and K12309 support the result of the structure comparison and twilight values. Although both sequences are still present with a shared evolutionary history and a recent common ancestor, their divergence present is still recognizable for the alignment and phylogeny algorithms.

A final note about the result of the K01220 who remains as a particular gene whose signal is unique and not related clearly with other sequences or groups. The prevalence of this signal could resemble their alternate function of *lacG* where the recognition of the phosphorylated galactoside substrate could bring further distinctive elements involved and present in the structure and even in their primary sequence. Is in a recent study of Zhong *et al.* [38] where this particular gene is reported as an element suitable to interact with microbial communities, these

kind of genes although diverse, their signal and their divergence sequences are not presented. The prevalence of particular motifs in this gene and structural regions could be the key to interact with different metabolism in alternate species among certain communities.

Conclusions

The development ~~a~~ method to compare data from orthologous groups, coupling them with specific crystal structures enable to have an additional features who complement the homology proposals such as combining with functional recognition of motifs and even in later stance to complement the arrangement with and the particular phylogenetic signal that could have each orthologous group. The alignment and phylogeny using previously reported orthologous.

A simple question using β-galatosidases and primary *lacZ* gene make possible to define that a classic report where *ebgA* remains as homologous from lacZ now this proposal has a deep relation with its feature, their structural homology and their functional separation of domains.

The recognition of a simple question about the diversity of a single degrading reaction into a set of genes enable to recognize that although the selection of this set of sequences, the homology obtained in at least two couples of KO orthologous groups bring the possibility to make further research and establish a particular set of relation among this KO groups. A differential study where Kooning *et al* [2] have tried to relate only in bioinformational way a set of conserved genes need to be taken as reference. As the study show, is necessary to combine certain signal functional elements and even more *in silico* essay to only evaluate if is possible to overpass the twilight results of an structural comparison into a plausible homology that still is not proposed and might include a particular set of gene families [39]

The particular effort to recognize how this particular reaction is diverse in orthologous genes can be a key to explore how these reactions could solve and be the element who can resolve since evolutionary questions to biotechnological needs in the future.

KEGG KO group	Definition	EC number	Reaction	Gene-related Group	PDB code	Structure	Obtained from organism
K12111	Evolved beta-galactosidase subunit alpha	3.2.1.23	Hydrolysis of terminal non reducing beta-D-galactose residues in beta-D-galactosides	*ebgA*	6S6Z	Octamer	*Thermotoga maritima*
K01190	Beta galactosidase	3.2.1.23	Same as K12111	*lacZ, gly2E, gal2C, rhgZ, yesZ*	3J7H	Tetramer	*Escherichia coli*-K12
K12309	Beta galactosidase	3.2.1.23	Same as K12111	*GLB1*	6S6Z	Octamer	*Homo sapiens*
K25543	Beta-galactosidase-1-like-protein	3.2.1.-	unknown	*GLB1L*	4MAD	Dimer	*Niallia circulans*
K12308	Beta galactosidase	3.2.1.23	Same as 12111	*bgaB, lacA, bglY*	5E9A	Pentamer	*Rahnella* sp. R3
K18579	Galactan endo-1, 6-beta-galactosidase	3.2.1.164	Endohydrolysis of (1→6)-beta-D-galactosidic linkages in arabinogalactan proteins and (1→3):(1→6)-beta-galactans to yield galactose and (1→6)-beta-galactobiose as the final products	*6GAL*	6IUJ	Dimer	*Talaromyces cellulolyticus*
K01224	Arabino-galactan endo-1-4-beta-galactosidase	3.2.1.89	Hydrolyses the beta(1→4) bonds found in homologous from plants. Produces galactan endo-beta-1,3-galactanase	-	1R8L	Dimer	*Bacillus licheniformis*
K01220	6 Phospho-beta-galactosidase	3.2.1.85	6-phospho-beta-D-galactoside + H_2O → 6-phospho-D-galactose + alcohol	*lacG*	1PBG	Dimer	*Lactococcus lactis*
K12112	Evolved beta-galactosidase subunit beta	-	unknown	*ebgC*	-	-	-

Table 1: KO groups analyzed in this study. The orthologous groups are described with all the data from KEGG and their gene denomination where is present. The corresponding PDB code is obtained after recognize a particular similar sequence from the database compared with the consensus sequence from the alignment

KEGG KO group	Pfam motif recognized	Features (obtained e-value)
K12111	Glyco hydro 2 N Glyco hydro 2 Glyco hydro 2 C	Glycosyl hydrolases family 2, sugar binding domain (1e-61) Glycosyl hydrolases family 2 (2.3e-14) Glycosyl hydrolases family 2, TIM barrel domain (2e-118)
K01190	Glyco hydro 2 N Glyco hydro 2 Glyco hydro 2 C lacZ 4 Bgal small N	Glycosyl hydrolases family 2, sugar binding domain (1.5e-46) Glycosyl hydrolases family 2 (2.1e-19) Glycosyl hydrolases family 2, TIM barrel domain (7.3e-122) Not defined (6.7e-20) Beta galactosidase small chain (4.1e-58)
K12309	Glyco hydro 35	Glycosyl hydrolases family 35 (3.2e-123)
K25543	Glyco hydro 35	Glycosyl hydrolases family 35 (4.1e-112)
K12308	Glyco hydro 42 Glyco hydro 42M	Beta-galactosidase (2e-154) Beta-galactosidase trimerisation domain (2e-75)
K18579	Glyco hydr 30 2	O-Glycosyl hydrolase family 30 (1.3e-16)
K01224	Glyco hydro 53	Glycosyl hydrolase family 53 (7.3e-123)
K01220	Glyco hydro 1	Glycosyl hydrolase family 1 (1.6e-166)

Table 2: Pfam motifs recognized by the FASTA sequence of the crystal structure and the conserved sequence from the corresponding KO group. The numbers and description where extracted from KEGGweb page from particular searches [20]

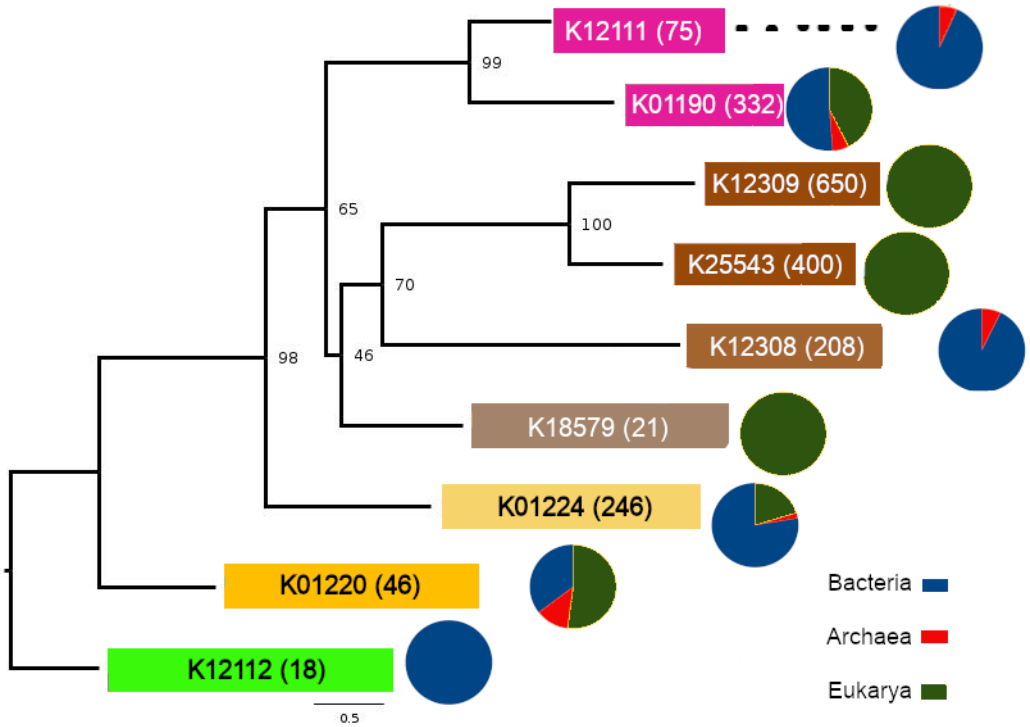

Figure 1: Maximum – likelihood phylogenetic tree using Bacterial and Eukaryal conserved sequences from each KO groups of β-galactosidases. The branch names are presented with the corresponding KO group and all the genera counted in parenthesis in white and black letters and using a color code to detect and sign their relation with later figures. Close to the branch name is presented a pie chart where are represented elements found from the three cellular domains, its color guide is presented in the lower right of the figure. The recognized bootstrap for these phylogeny is presented in the nodes. The distance scale is presented in the lower left of the figure

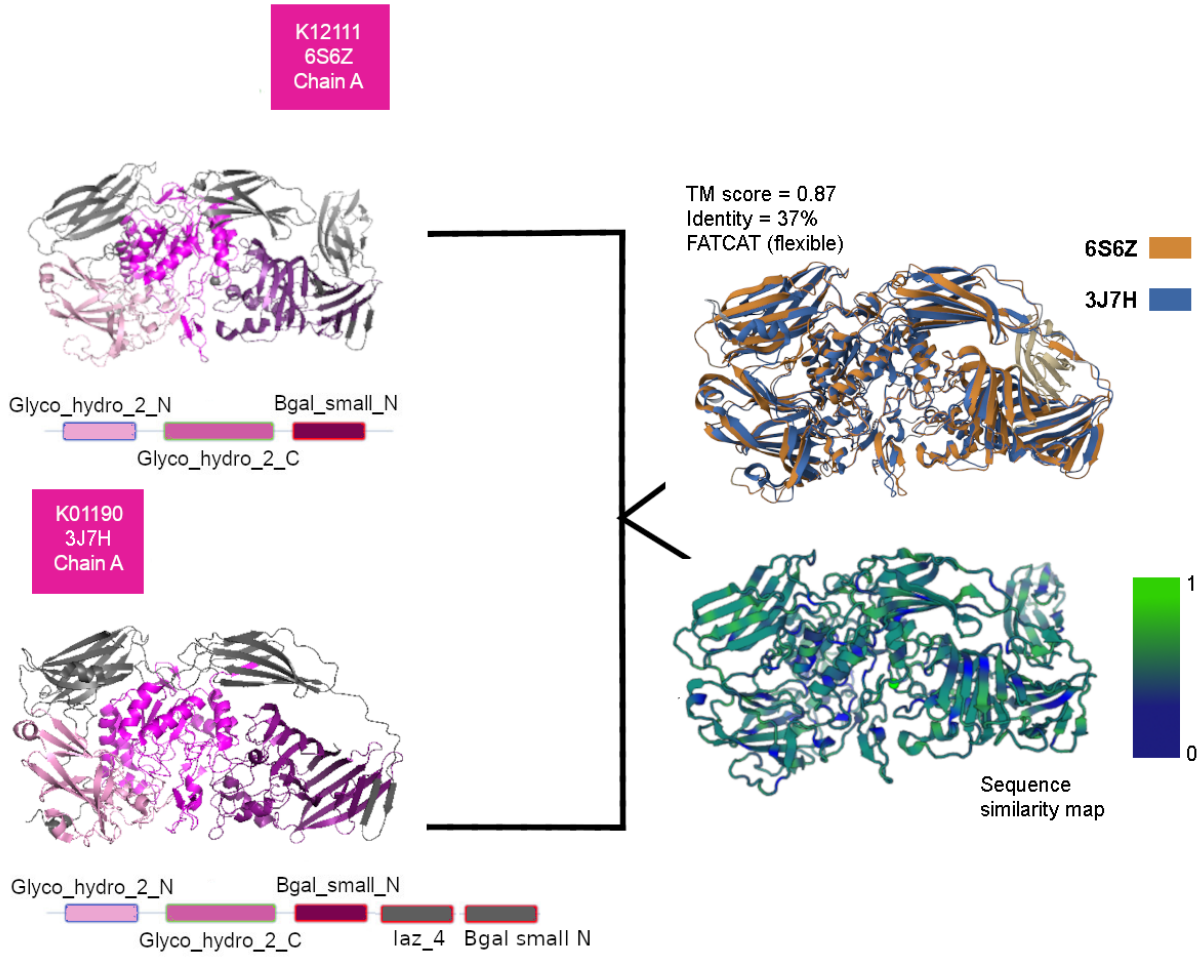

Figure 2: Comparison of crystal protein structures representing K1211 and K01190 orthologous groups. The Pfam recognized motifs are represented with similar color of the KO group in each tertiary structure. Both sequences were aligned and analyzed in PDB page. The results obtained from the comparison are presented in the top left of the alignment by Sequences similarity map obtained from Dali is presented in the lower part of the scheme

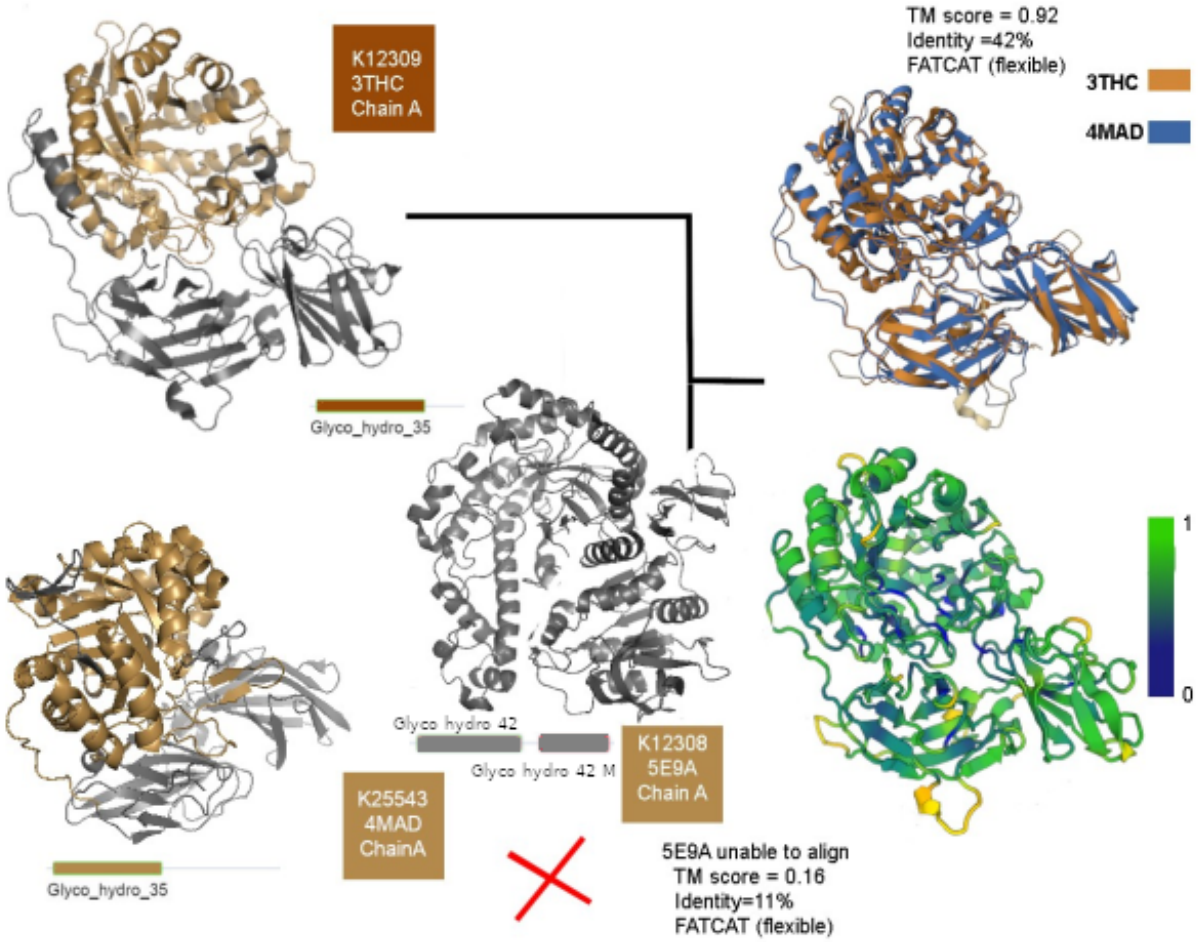

Figure 3: Comparison of crystal protein structures representing K12308, K12309 and K25543 orthologous groups. The Pfam recognized motifs are represented with similar color of the KO group in each tertiary structure. For this case, the comparison among the three crystal corresponding to a specific orthologous group is presented as the same folding with the orthologous color pattern and in gray for the crystal structure that is incompatible and non homologous with the compared folding. The values of each comparison are located near to the crystals and the tertiary structure alignment and For this case, this comparison enable to avoid one crystal from the homologous arrangement

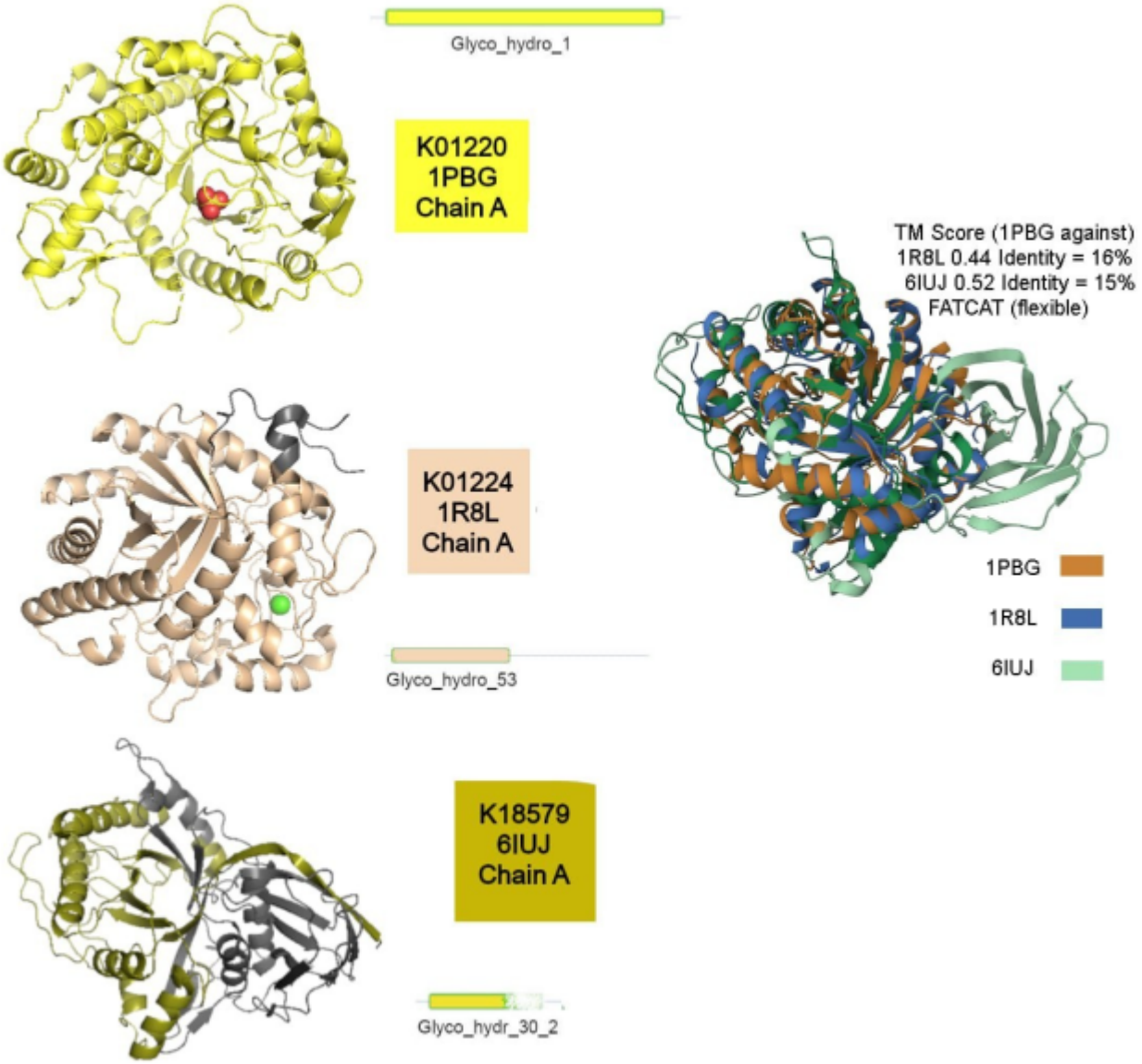

Figure 4: Comparison of crystal protein structures representing K01220, K01224 and K18579 orthologous groups. The Pfam recognized motifs are represented with similar color of the KO group in each tertiary structure. The structural alignment is presented with TM values and color pattern to differentiate the three optimal homologous.

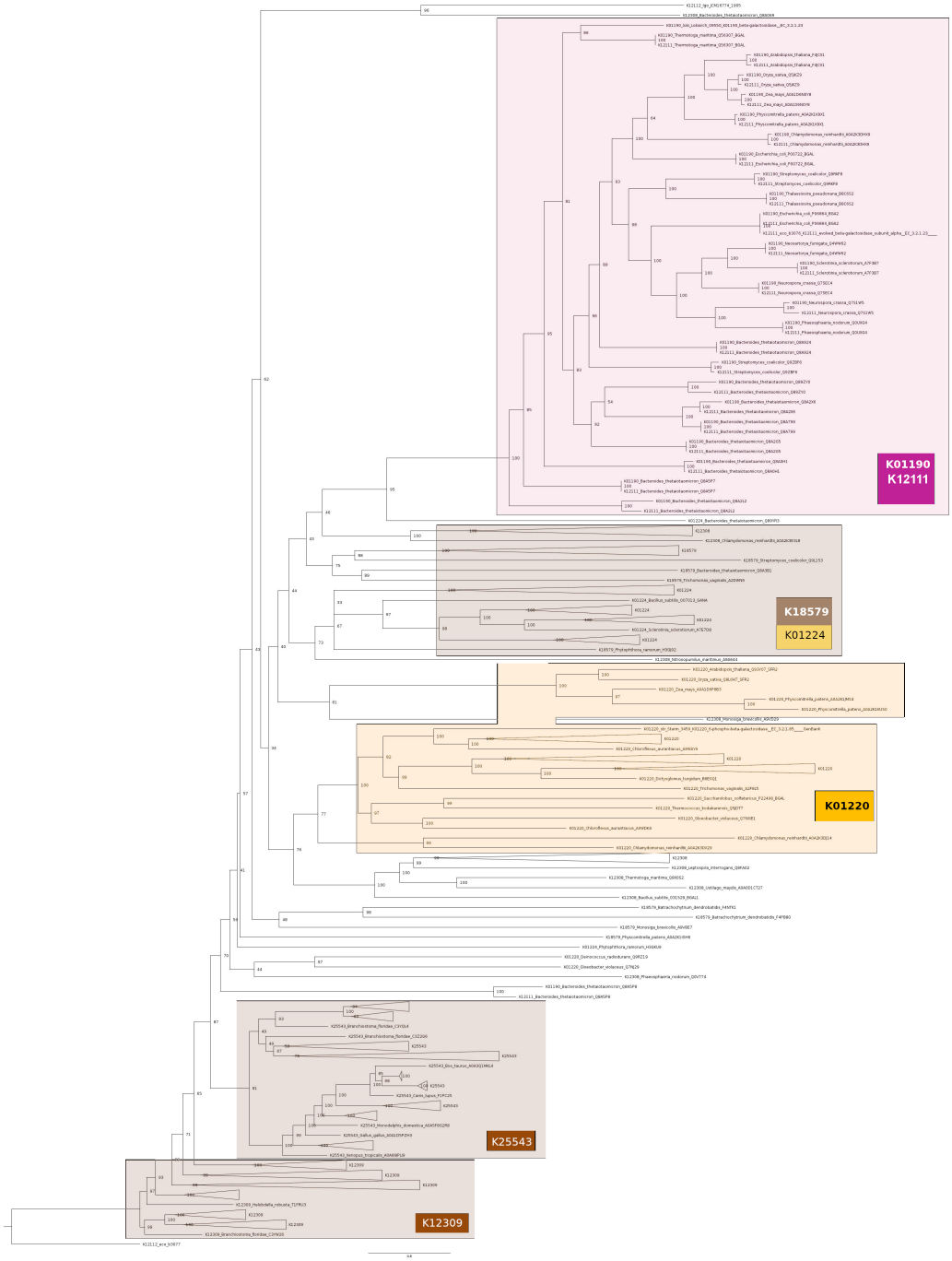

Figure 5: Figure 4. Maximum likelihood phylogenetic tree integrating all the revised orthologous from the SHOOT.bio databases coupled with the KEGG KO groups, this figure can point those groups whose similarity make them cluster into adjoining branches.Each triangle is a collapsed group of homologous from the same KO group where the simplification is originated if the KO groups is the same. The shadow is only an effect to link the KO groups presented

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